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Research Topic: *Use of Bioluminescent Bacteria in Detection of Electromagnetic Wave Pollution*

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ABSTRACT

Electronic devices and ultraviolet (UV) radiation from the sun are factors that lead to the propagation of electromagnetic waves. With current technology, it is not possible to prevent the propagation of electromagnetic waves. However, the uncontrolled increase in the quantity of electromagnetic wave propagation, particularly at high levels, leads to electromagnetic wave pollution. Electromagnetic pollution causes various health and psychological problems in humans, disrupts animal migration routes, damages agricultural areas, and results in ecological imbalances. In this context, it is crucial to rapidly identify regions with high electromagnetic wave pollution to implement preventive measures. Currently, electromagnetic wave pollution is detected using electronic devices that measure electromagnetic waves. However, these devices are costly and require calibration. This can make the process of measuring electromagnetic pollution challenging at times. In this regard, the study explores the feasibility of using bioluminescent bacteria for the rapid and accurate detection of electromagnetic wave pollution. After identifying the appropriate bioluminescent bacteria, their light emission will be evaluated in response to electromagnetic wave exposure.

GOAL

This research proposal presents a biotechnological solution for the detection of electromagnetic wave pollution. The aim of this study is to develop a method that is fast, consistent, cost-effective, and efficient for the detection of electromagnetic pollution. A literature review revealed no previous studies utilizing bioluminescent bacteria for the detection of electromagnetic wave pollution. This finding supports the innovative aspect of the proposed study. Additionally, since this proposal represents a novel approach in this field, it is expected to contribute significantly to the existing body of literature.

INTRODUCTION

Some organisms emit visible light as a result of oxidative reactions that occur within their bodies, releasing chemical energy. This phenomenon is defined as bioluminescence, and the emitted light is referred to as bioluminescent radiation. Organisms capable of bioluminescent emission can generate this light through the synthesis of luciferase enzymes (a general term for oxidative enzymes involved in the bioluminescence process) and the interaction of the relevant substrate [1].

The bioluminescence process is associated with oxidation. The luciferase enzymes present in the cells of bioluminescent organisms, in the presence of luciferin, ATP, and oxygen, produce bioluminescent radiation [2]. The reactions involved in the bioluminescence process are presented in Figure 1.

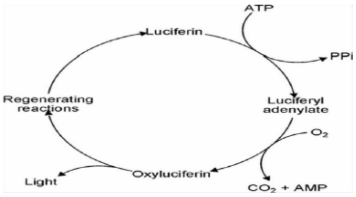


Figure 1. *Bioluminescent Reactions [2, p. 19].*

Currently, bioluminescent organisms are utilized in both in-vivo and in-vitro applications such as cancer and infection tracking, stem cell monitoring, protein localization, and protein interaction analysis. Bioluminescence-based imaging stands out due to its high sensitivity, ease of use, cost-effectiveness, consistency, efficiency, and practicality [1].

Electromagnetic waves are a type of wave that originates from a radio frequency source and propagates through both vacuum and material media [3].

Human-made and natural factors such as base stations, satellite systems, radio, television, Wi-Fi, and ultraviolet (UV) waves reaching the Earth in excess due to ozone layer

depletion contribute to electromagnetic wave pollution. This phenomenon negatively affects living organisms and leads to environmental degradation. Moreover, electromagnetic pollution can impair the efficiency and functionality of avionics systems used in the aviation industry [4].

The World Health Organization (WHO), the International Commission on Non-Ionizing Radiation Protection (ICNIRP), and the Institute of Electrical and Electronics Engineers (IEEE) have established exposure limits to protect individuals from excessive electromagnetic radiation. Within this framework, the exposure limit has been set at 41.25 V/m for 900 MHz and 58.33 V/m for 1800 MHz. Exceeding these thresholds may result in adverse health effects. Therefore, the measurement and mapping of electromagnetic wave pollution are of great importance for the implementation of preventative measures [5]

Today, electromagnetic pollution is typically detected using automated or computer-controlled devices. However, the high cost and occasional need for calibration of high-quality electromagnetic pollution measuring instruments can render the monitoring process challenging [6]

RESEARCH QUESTION

Can bioluminescent bacteria be used to detect electromagnetic wave pollution through measurable changes in bioluminescence intensity?

PROPOSED METHODOLOGY

The proposed experimental methodology in this study is presented in three phases. In the first phase, bacterial strains with bioluminescent properties reported in the literature (such as Vibrio, Photobacterium, Shewanella (Alteromonas), and Photorhabdus (Xenorhabdus)) will be identified. Following identification, the selected bacterial strains will be exposed to electromagnetic waves with wavelengths starting from 380 nm and decreasing, under controlled conditions including a temperature range of 20 °C to 30 °C, suitable culture media, appropriate humidity, and pH levels. If one or more of these strains exhibit light emission behavior in response to this exposure, they will be isolated. The light emission of the bacteria will be measured and analyzed using high-sensitivity spectrophotometric devices both before and after electromagnetic exposure [7].

In the second phase, the bacterial strains that exhibited changes in light intensity will be placed in environments identified as having high levels of electromagnetic wave pollution based on current standards. In these settings, the light emission of the selected bioluminescent bacteria will be monitored. In cases where light emission is observed, the intensity of the emitted light will be quantitatively measured and analyzed. To ensure the reliability of measurements obtained from the bioluminescent bacteria, simultaneous electronic measurements will also be conducted, and the resulting data will be statistically compared.

In the third phase, bioluminescent bacterial strains that demonstrate sensitivity to specific types of electromagnetic radiation will be evaluated in terms of availability, cost-efficiency, and practical applicability.

The methodology presented in this study is proposed as a preliminary framework and does not represent a completed research study. The methodology may be further detailed and modified upon implementation of the study.

EXPECTED RESULTS

The proposed study aims to investigate whether bioluminescent bacteria can be utilized for the detection of electromagnetic wave pollution. In this context, the research seeks to develop an alternative measurement method. It is anticipated that, as a result of experimental studies, certain species of bioluminescent bacteria will exhibit sensitivity to harmful electromagnetic waves, particularly those with wavelengths shorter than 380 nm. Accordingly, the study aspires to create a biotechnological, practical, and innovative electromagnetic wave indicator.

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